Text Mining 2026

Lecture 5: Named entity detection and classification

Ilia Markov

What is an entity?

Instances of people, organisations, places, objects, incidents that exist in some world

What is an entity?

What is reference?

Instances of people, organisations, places, objects, incidents that exist in some world

What is an entity?

A communicative act to identify an entity (point at someone, picture identifying someone)

What is reference?

Text mining: referring expressions in a language

What is a referring expression?

Instances of people, organisations, places, objects, incidents that exist in some world

What is an entity?

A communicative act to identify an entity

What is reference?

Name or proper noun, common noun phrase or pronoun,..

What is a referring expression?

but in the case of events and properties also verbs, verb phrases, adjectives etc….

What is an entity?

Instances of people, organisations, places, objects, incidents that exist in some world

What is reference?

A communicative act to identify an entity

What is a referring expression?

Proper noun, common noun phrase or pronoun,..

What is a named entity expression?

Definite noun phrases, proper nouns

What is Named Entity Recognition?

- Named Entity Recognition is the task of finding and classifying **named entity expressions** in text.

- E.g., text about Abraham Lincoln:

Abraham Lincoln Listeni/ˈeɪbrəhæm ˈlɪŋkən/ (February 12, 1809 – April 15, 1865) was the 16th President of the United States, serving from March 1861 until his assassination in April 1865. Lincoln led the United States through its Civil War—its bloodiest war and its greatest moral, constitutional and political crisis.[1][2] In doing so, he preserved the Union, abolished slavery, strengthened the federal government, and modernized the economy.

Lincoln grew up on the western frontier in Kentucky and Indiana. Largely self-educated, he became a lawyer in Illinois, a Whig Party leader, and a member of the Illinois House of Representatives, where he served from 1834 to 1846. Elected to the United States House of Representatives in 1846, Lincoln promoted rapid modernization of the economy through banks, tariffs, and railroads. Because he had originally agreed not to run for a second term in Congress, and his opposition to the Mexican–American War was unpopular among Illinois voters, Lincoln returned to Springfield and resumed his successful law practice. Reentering politics in 1854, he became a leader in building the new Republican Party, which had a statewide majority in Illinois. In 1858, while taking part in a series of highly publicized debates with his opponent and rival, Democrat Stephen A. Douglas, Lincoln spoke out against the expansion of slavery, but lost the U.S. Senate race to Douglas.

source: http://en.wikipedia.org/wiki/Abraham_Lincoln

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People

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People

Locations

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People

Locations

Organisations

- All these things are in the world in a particular time and place -> they exist.
- Republican party -> the members can change, but the entity as an organization remains.

source: http://en.wikipedia.org/wiki/Abraham_Lincoln

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People

Time

Locations

Organisations

- Time can be an entity: point to specific dates, periods (e.g., from March)
- No agreement on what entities should be distinguished and be part of the task

source: http://en.wikipedia.org/wiki/Abraham_Lincoln

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People

Time

Locations

Events

Organisations

- ‘Civil War’ - named events (famous)
- ‘Traveling to work’ - exists but no name given

source: http://en.wikipedia.org/wiki/Abraham_Lincoln

Named entity detection and linking (NERC-D/L)

- NER(Recognition): detecting the phrase that is the name of an entity
- NEC(Classification): assigning an entity type to the phrase (person, location..)
- NEL(Linking) or NED(Disambiguation): establishing the identity of the entity in a given reference database (Wikipedia, DBPedia, YAGO)
- Coreference: any phrase that makes reference to an entity instance, including pronouns, noun phrases, abbreviations, acronyms, etc.
- NER and NEC - pipeline solutions:

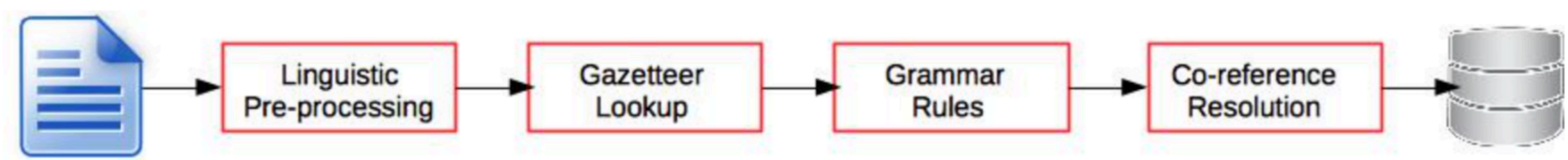
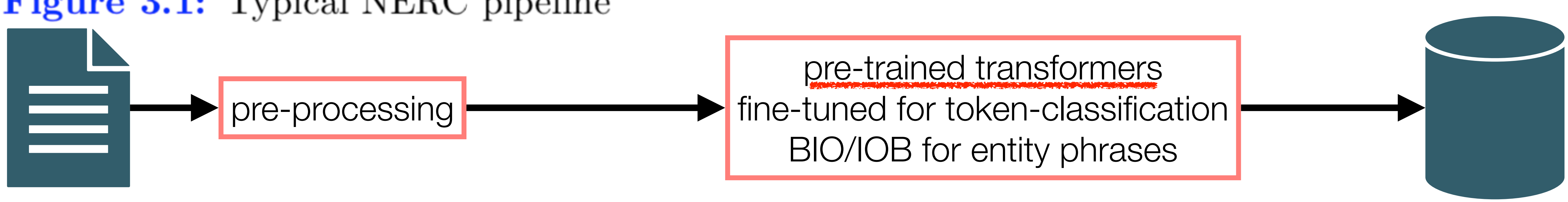


Figure 3.1: Typical NERC pipeline



What makes it a hard task?

- **Variation:** IBM, The Big Blue; New York, NY, The Big Apple (more popular -> more names)
- **Ambiguity**:
  - *may may still rule in may.* *(may -> name (lowercased, social media), noun (month), modal verb)*
  - *Austin Reed (fashion shop), Parkinson's disease, Pythagoras' Theorem*
  - *Miami Vice (tv series), Brazil (movie)*
- **Extent**: *Sir Robert Walpole; [Abraham$_{B}$ Lincoln$_{I}$,$_{O}$ [the$_{O/B}$ 16th$_{O/I}$ President$_{B/I}$ of$_{I}$ the$_{I}$ [United$_{B/I}$ States$_{I}$]]]* <— nested entities (whole phrase vs. 3 entities; low IAA - borders)
- **Types**: price, age, PER or also social roles of people: *professions, criminal/victim, doctor, patient, teacher, student*
  - http://nerd.eurecom.fr/ontology (classes / types of NEs)
  - Fine-grained entity typing (e.g., FIGER uses 112 types from Freebase)
- **Time**: 8-3-2022, Tuesday, 8am, yesterday (relative expressions), this week
- **Metonymy**: US, Holland, The Netherlands, Mexico, Ford —> people, organisation, location

NERC feature engineering

- Word-level features (structural, semantic, …)
- Lookup features (lists, databases)
- Document & Corpus features
  (coherence: news article about the US in a particular period will involve the same people or organizations that are dominant in that period)

Word level features

Table 1. Word-level features

Features	Examples
Case	– Starts with a capital letter – Word is all uppercased – The word is mixed case (e.g., ProSys, eBay)
Punctuation	– Ends with period, has internal period (e.g., St., I.B.M.) – Internal apostrophe, hyphen or ampersand (e.g., O'Connor)
Digit	– Digit pattern (*see Section 3.1.1*) – Cardinal and ordinal – Roman number – Word with digits (e.g., W3C, 3M)
Character	– Possessive mark, first person pronoun – Greek letters
Morphology	– Prefix, suffix, singular version, stem – Common ending (*see Section 3.1.2*)
Part-of-speech	– proper name, verb, noun, foreign word
Function	– Alpha, non-alpha, n-gram (*see Section 3.1.3*) – lowercase, uppercase version – pattern, summarized pattern (*see Section 3.1.4*) – token length, phrase length

word shape:
- Xxx, Joe,
  Xxxxxxxxx, Amsterdam,
  xxxXx spaCy
- dd-dd-dddd, 12-10-2007

short word shape:
- Xx, Joe,
  Xx, Amsterdam,
  xXx spaCy

From Nadeau, D., & Sekine, S. (2007)

Gazetteers (lists) & lexicons

Table 2. List lookup features.

Features	Examples
General list	– General dictionary (see Section 3.2.1) – Stop words (function words) – Capitalized nouns (e.g., January, Monday) – Common abbreviations
List of entities	– Organization, government, airline, educational – First name, last name, celebrity – Astral body, continent, country, state, city
List of entity cues	– Typical words in organization (see 3.2.2) – Person title, name prefix, post-nominal letters – Location typical word, cardinal point

From Nadeau, D., & Sekine, S. (2007)

Document features

Table 3. Features from documents.

Features	Examples
Multiple occurrences	– Other entities in the context – Uppercased and lowercased occurrences (see 3.3.1) – Anaphora, coreference (see 3.3.2)
Local syntax	– Enumeration, apposition – Position in sentence, in paragraph, and in document
Meta information	– Uri, email header, XML section, (see Section 3.3.3) – Bulleted/numbered lists, tables, figures
Corpus frequency	– Word and phrase frequency – Co-occurrences – Multiword unit permanency (see 3.3.4)

From Nadeau, D., & Sekine, S. (2007)

NERC data sample: features for token-level classification

https://www.kaggle.com/abhinavwalia95/entity-annotated-corpus

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- id, lemma, **next**-lemma, **next-next**-lemma, **next-next**-pos, **next-next**-shape, **next-next**-word, **next**-pos, **next**-shape, **next**-word, pos, **prev**-iob, **prev**-lemma, **prev**-pos, **prev-prev**-iob, **prev-prev**-lemma, **prev-prev**-pos, **prev-prev**-shape, **prev-prev**-word, **prev**-shape, **prev**-word, sentence_idx, shape, word, tag

Feature ablation

- 0, thousand, of, demonstr, NNS, lowercase, demonstrators, IN, lowercase, of, NNS, __START1__, __start1__, __START1__, __START2__, __start2__, __START2__, wildcard, __START2__, wildcard, __START1__, 1, capitalized, Thousands, O
- 1, of, demonstr, have, VBP, lowercase, have, NNS, lowercase, demonstrators, IN, O, thousand, NNS, __START1__, __start1__, __START1__, wildcard, __START1__, capitalized, Thousands, 1, lowercase, of, O
- 2, demonstr, have, march, VBN, lowercase, marched, VBP, lowercase, have, NNS, O, of, IN, O, thousand, NNS, capitalized, Thousands, lowercase, of, 1, lowercase, demonstrators, O

CoNLL feature representation

Word	POS	Chunk	Short shape	Label
American	NNP	B-NP	Xx	B-ORG
Airlines	NNPS	I-NP	Xx	I-ORG
,	,	O	,	O
a	DT	B-NP	x	O
unit	NN	I-NP	x	O
of	IN	B-PP	x	O
AMR	NNP	B-NP	X	B-ORG
Corp.	NNP	I-NP	Xx.	I-ORG
,	,	O	,	O
immediately	RB	B-ADVP	x	O
matched	VBD	B-VP	x	O
the	DT	B-NP	x	O
move	NN	I-NP	x	O
,	,	O	,	O
spokesman	NN	B-NP	x	O
Tim	NNP	I-NP	Xx	B-PER
Wagner	NNP	I-NP	Xx	I-PER
said	VBD	B-VP	x	O
.	,	O	.	O

Sequence labelling

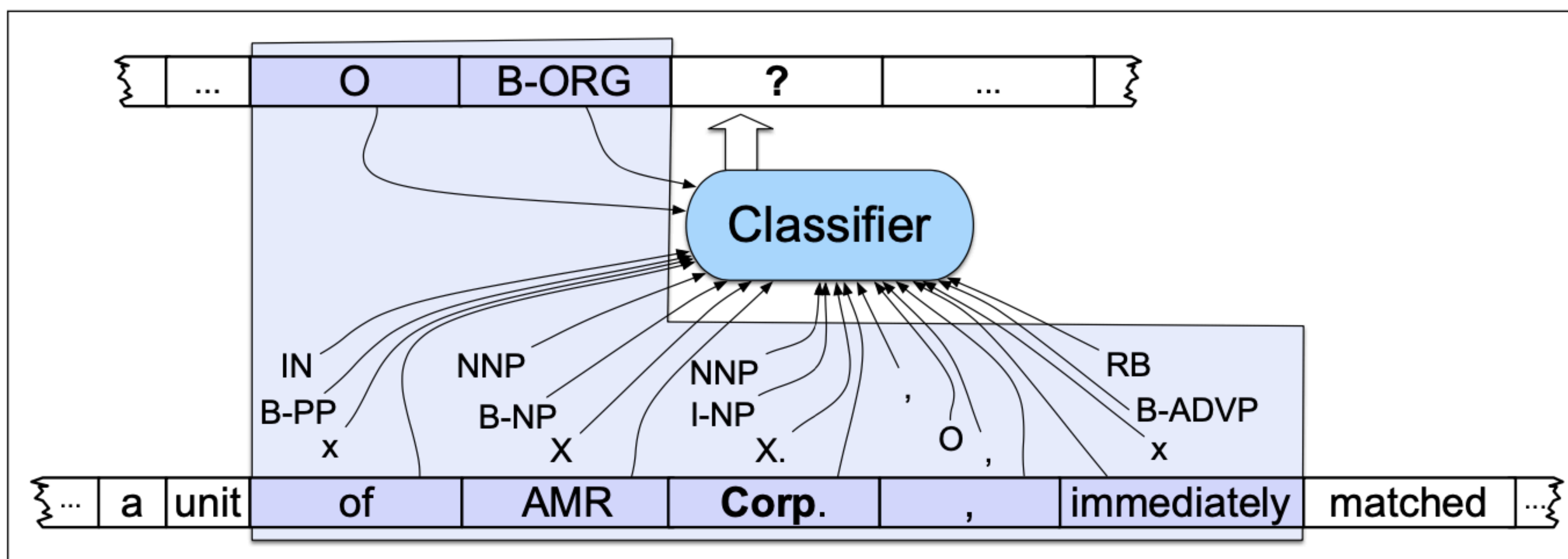
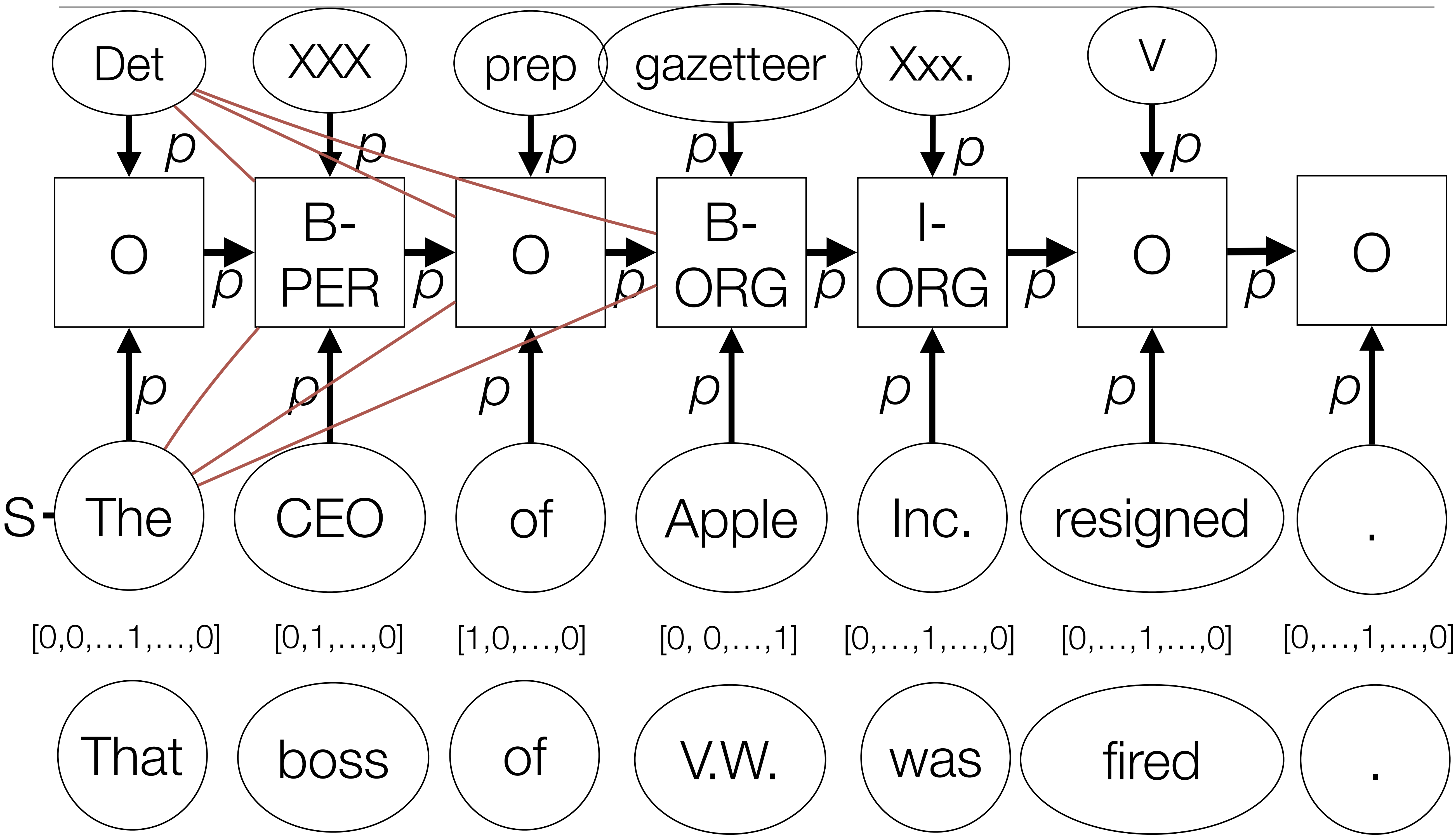


Figure 17.7 Named entity recognition as sequence labeling. The features available to the classifier during training and classification are those in the boxed area.

Juravsky & Martin 3ed, 2018

Conditional Random Field for IOB sequences



CRF: model sequence dependencies between tokens (left-to-right, right-to-left)

Sequence representation with one-hot-encodings & word embeddings

- **Feature space**
- vocabulary = [word-embedding]
- PoS = [A, NN, NNP, V]
- Dependencies = [nsubj, root, obj]
- Vector dimensions
  - [[300 embedding] + [0,0,0,0,0] + [0,0,0]]

(Combine embeddings with one-hot-encodings)

- [Vaccination, is, save]
- [NN, V, A]
- [nsubj, root, obj]

- [Vaccines, cause, autism]
- [NNP, V, NN]
- [nsubj, root, obj]

- [[.2,.1,.2,.5,1,.6,.1]
  [0,1,0, 0]
  [1,0,0]]
- [[.8,.1,.2,.5,1,.6,.1]
  [0,0,0,1]
  [0,1,0]]
- [[.1,.1,.1,.3,1,.2,.5]
  [1,0,0,0]
  [0,0,1]]

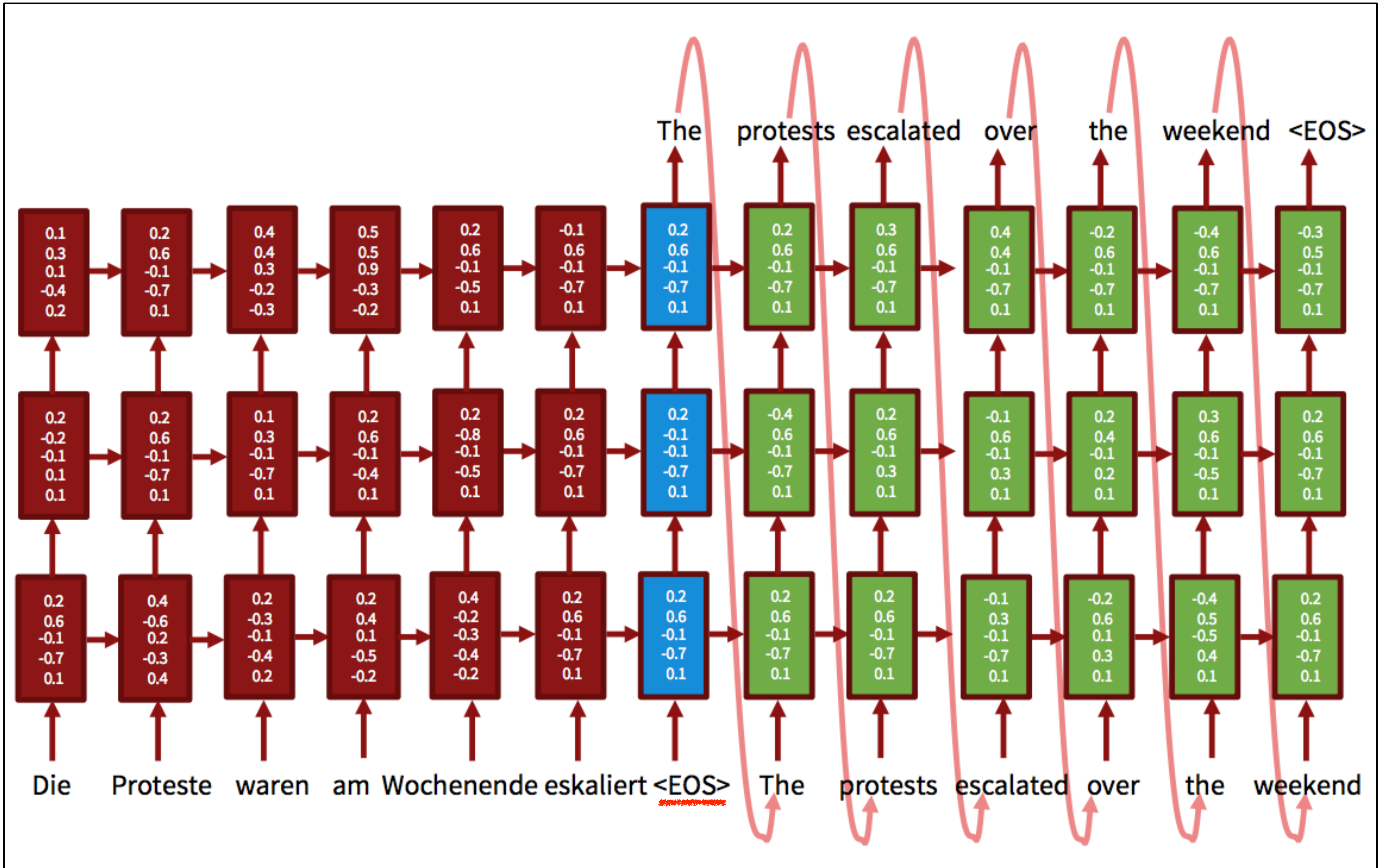
Token comparison

- [[.1,.1,.1,.3,1,.2,.4]
  [0,1,0,0]
  [1,0,0]]
- [[.3,.1,.2,.1,1,.3,.2]
  [0,0,0,1]
  [0,1,0]]
- [[.2,.1,.2,.5,1,.6,.1]
  [0,0,1,0]
  [0,0,1]]

- most similar to [.2,.1,.2,.5,1,.6,.1]
- medicine(s), antibiotics, penicillin, medication, insulin, treatment, injection, prescription, drug(s), therapy

Machine translation: long-short-term-memory

ENCODER
hidden layers

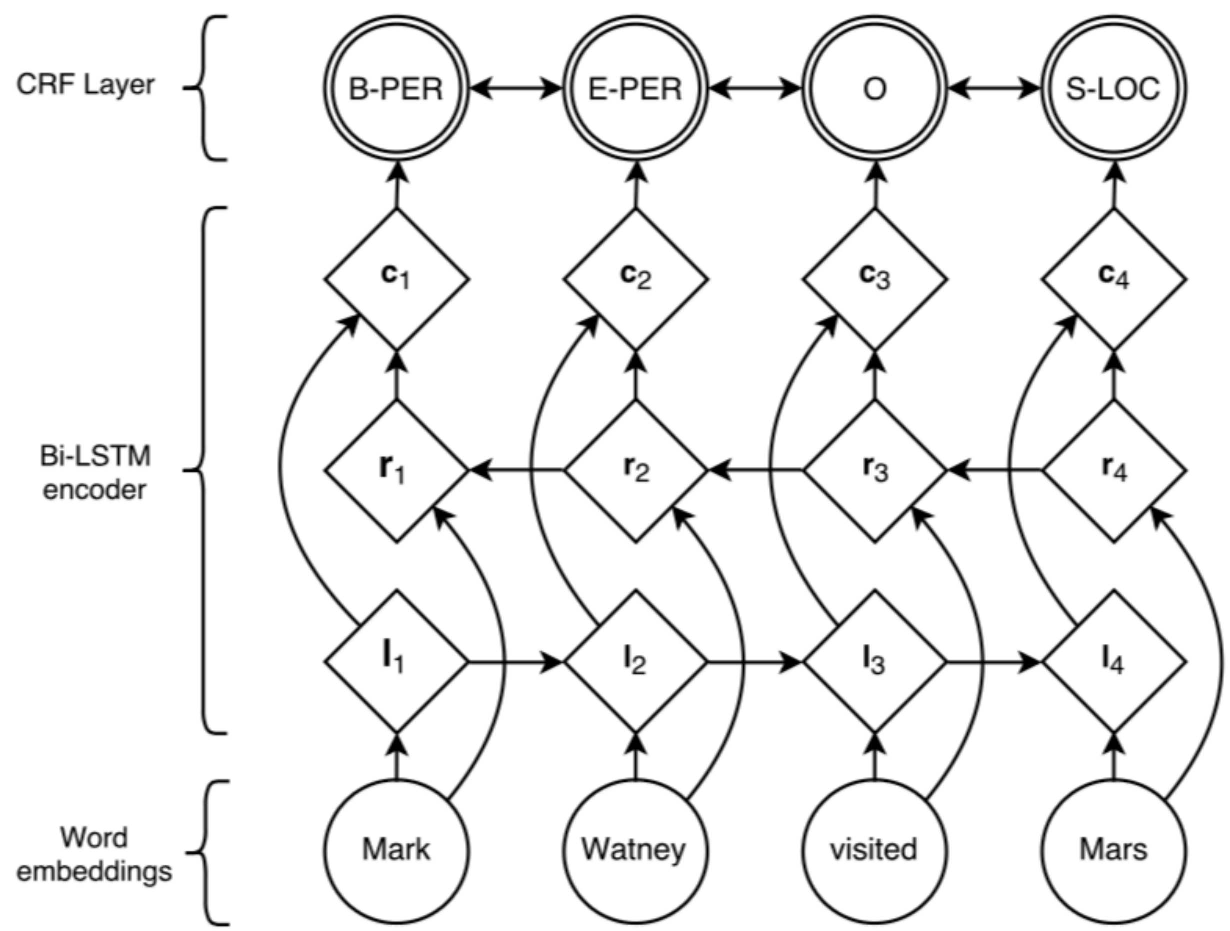


hidden layers
DECODER

Google Neural MT

Neural network (LSTM) and CRF

https://github.com/guillaumegenthial/tf_ner



Survey paper: Lample et al. (2016), Neural Architectures for Named Entity Recognition, NAACL.

Huang et al. (2015), Bidirectional LSTM-CRF Models for Sequence Tagging, arXiv.1508.01991v1

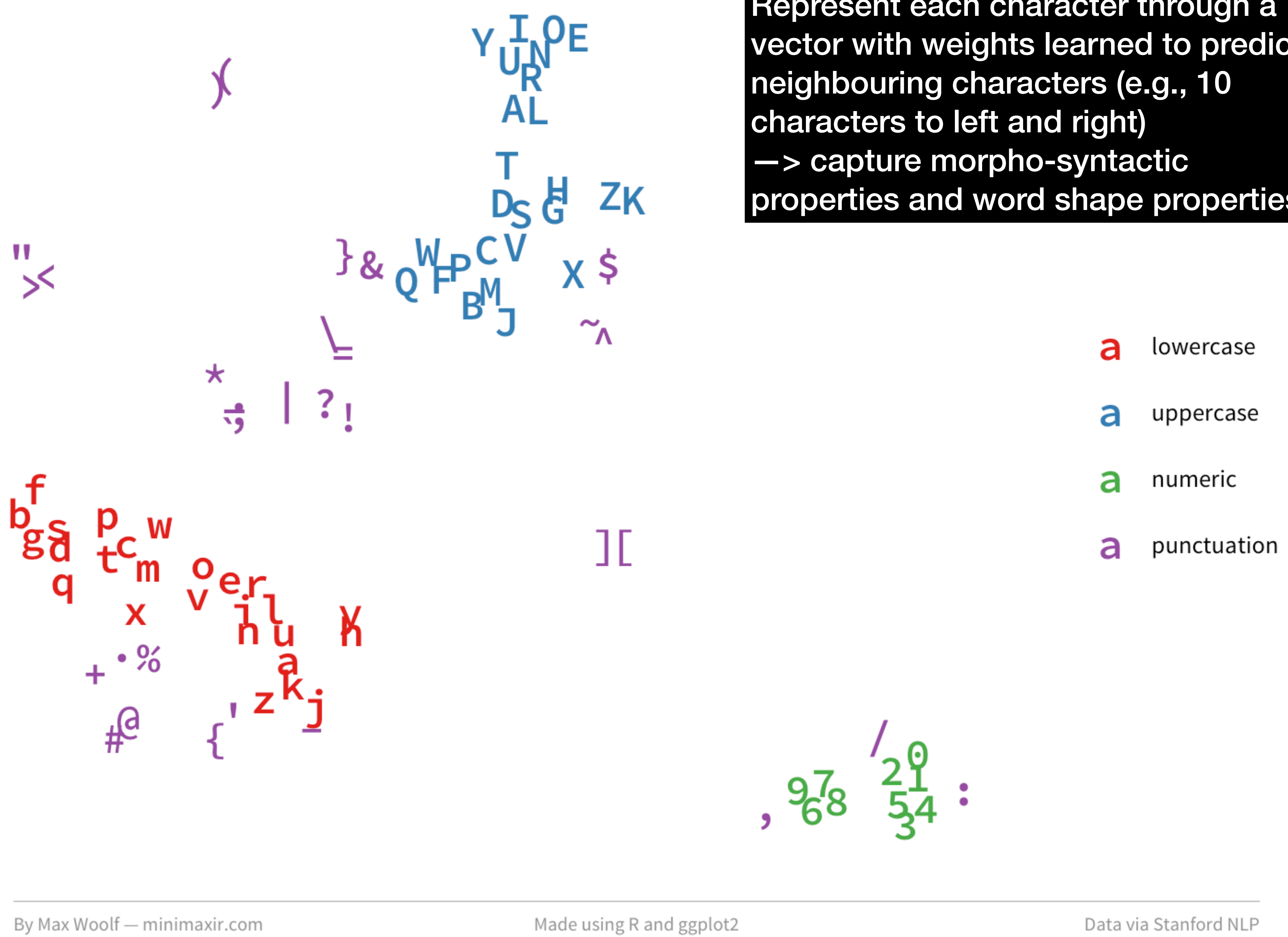
Long Short-Term Memory
LSTM

- Represent each word by its embedding representation
- BiLSTM encoder to represent these embeddings in a sequence (e.g., the embedding for ‘Watney’ is modified by preceded ‘Mark’): l layer -> left-to-right
- r layer -> same but moving right-to-left (we learn dependencies in 2 directions)
- Result is passed to c layer: 2-directional dependencies are combined
- Final token representation is used for CRF to do sequence token classification
- “-“ lose info about the actual word (e.g., word shape) -> character embeddings

Projection of 300D GloVe Character Vectors into 2D Space (16D, perplexity = 7)

Characters closer to each other are more similar in usage context.

Represent each character through a vector with weights learned to predict neighbouring characters (e.g., 10 characters to left and right)
—> capture morpho-syntactic properties and word shape properties



CRF on top of bi-LSTM using word & character embeddings

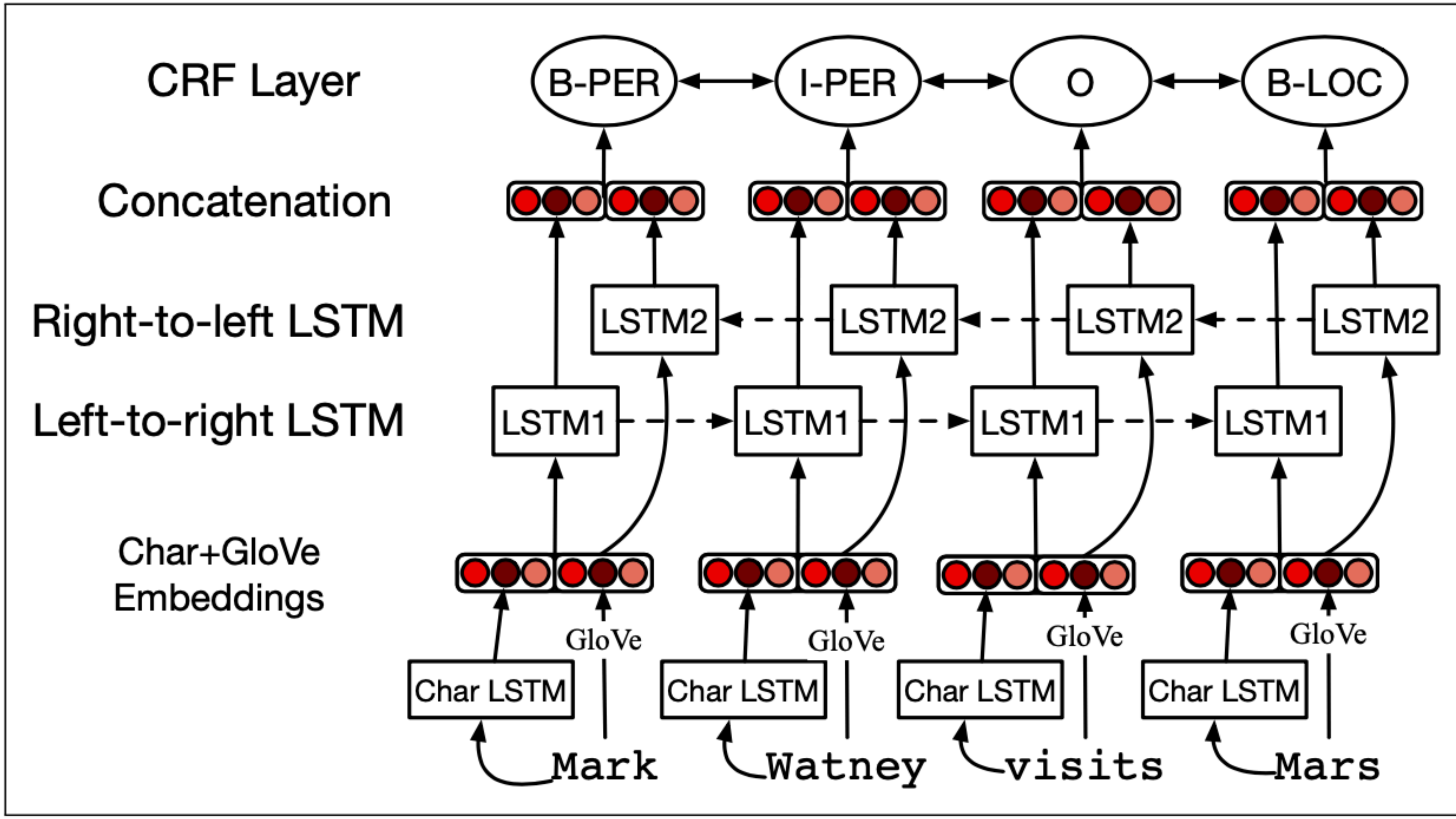


Figure 17.8 Putting it all together: character embeddings and words together a bi-LSTM sequence model. After (Lample et al., 2016)

- The embedding representation of the word ‘Mark’ is now extended with the output of the character LSTM model that represents the actual characters of the word ‘Mark’, but through embeddings
- Capture the structural properties through an embedding representation (char) and stick them to the semantic properties (words), and feed to the BiLSTM architecture to exploit the predictive value

Bidirectional LSTM models for NERC

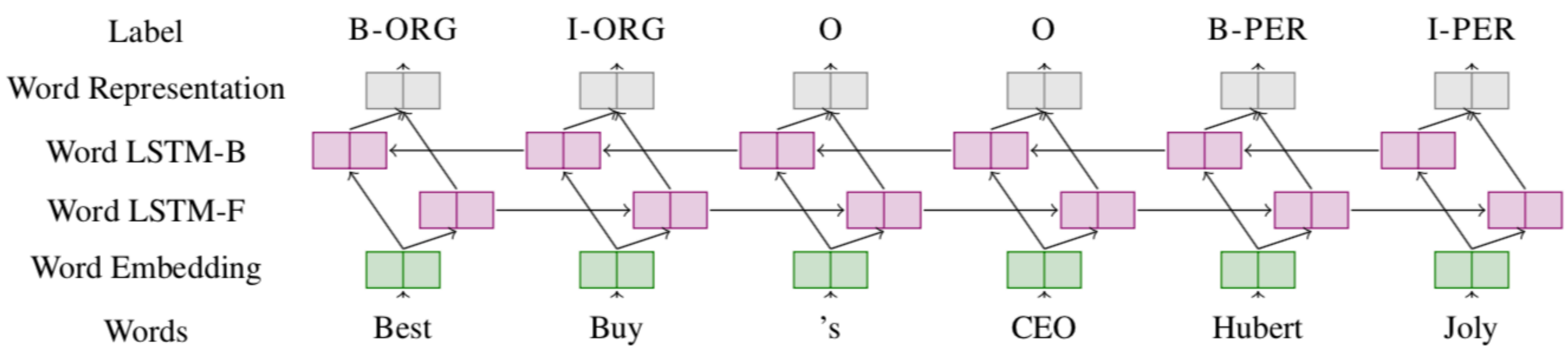


Figure 1: Word level NN architecture for NER

(only words replaced by word embeddings)

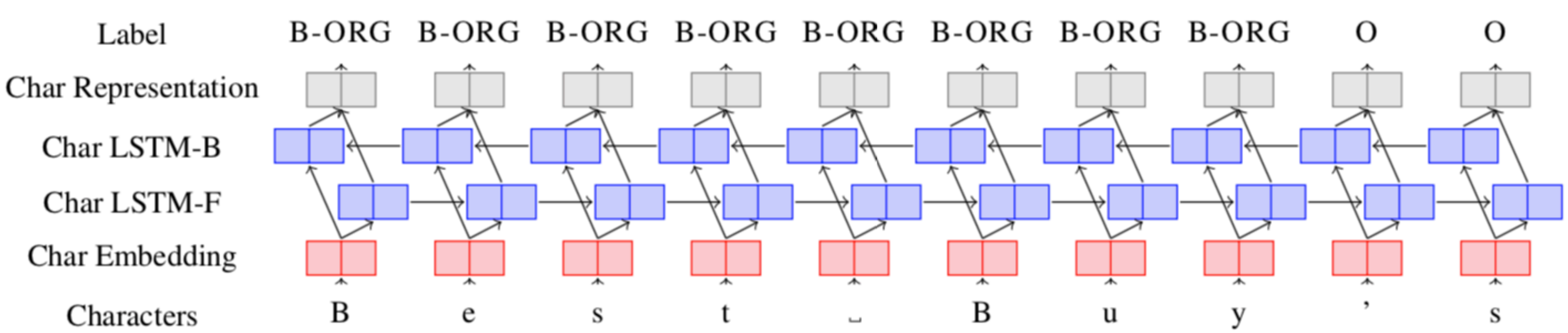


Figure 2: Character level NN architecture for NER

(only characters; structural but not semantic properties)

Yadav & Bethard 2018

https://towardsdatascience.com/besides-word-embedding-why-you-need-to-know-character-embedding-6096a34a3b10

Bidirectional LSTM models for NERC

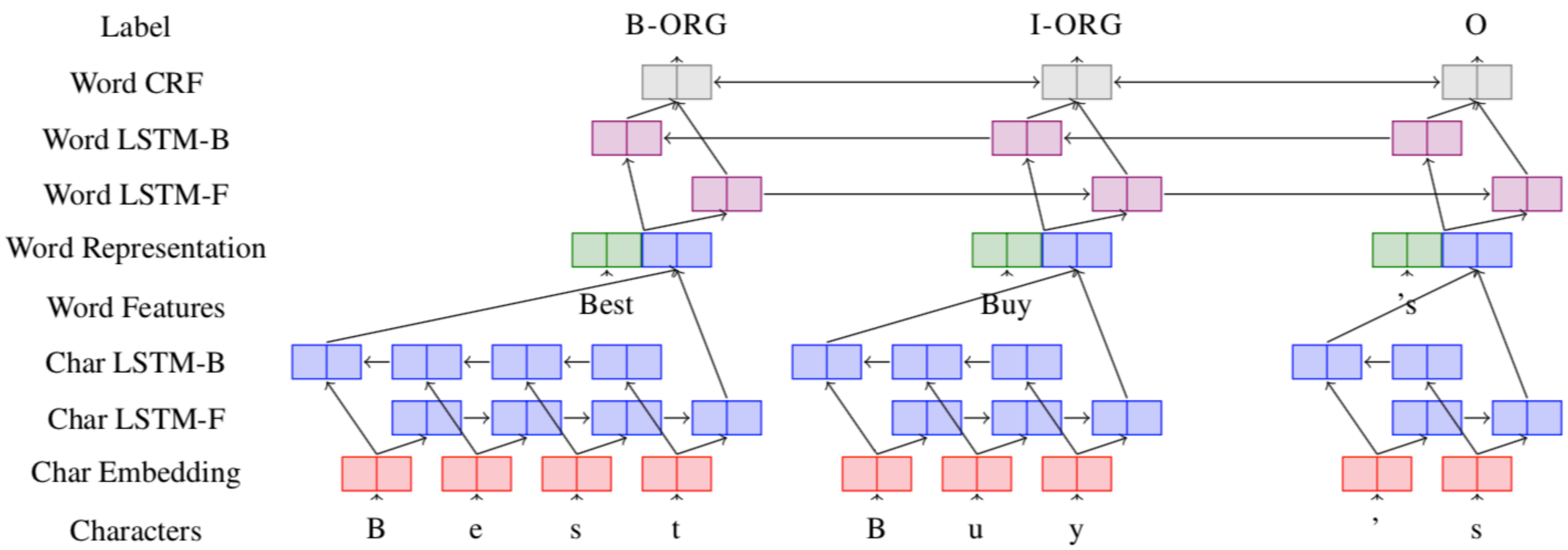


Figure 3: Word+character level NN architecture for NER

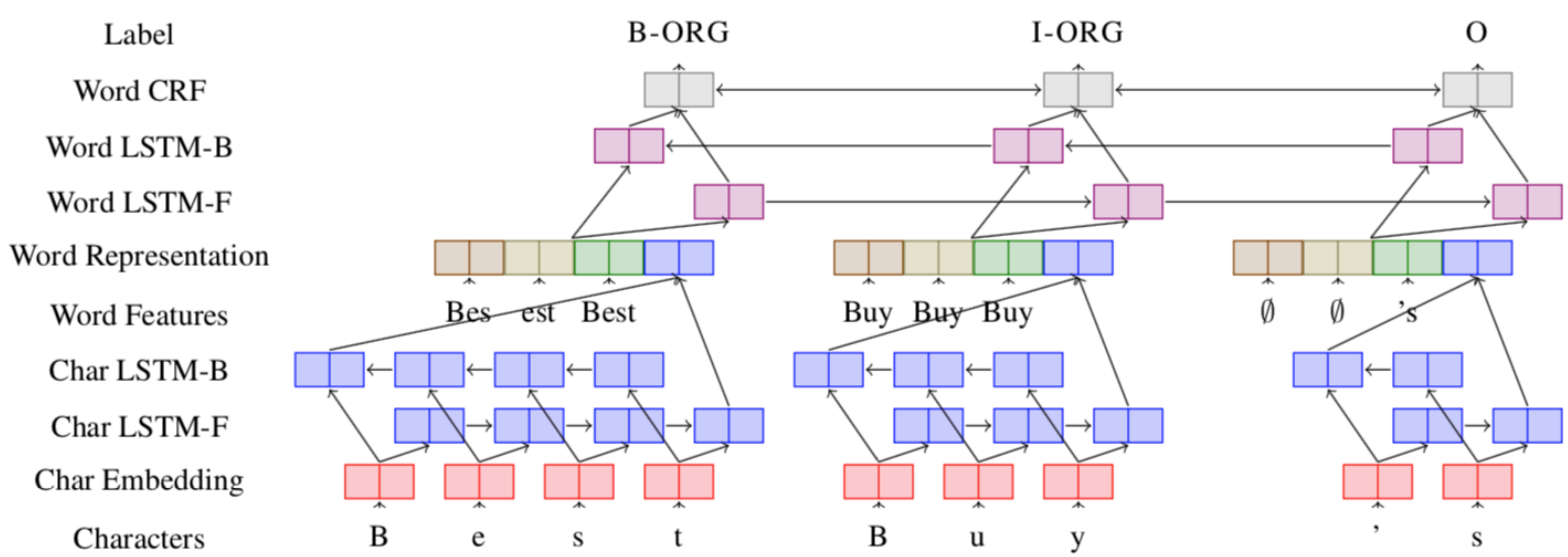


Figure 4: Word+character+affix level NN architecture for NER

Affix embeddings from all n-gram prefixes and suffixes of words in the training corpus to learn linguistic units rather than single characters

Yadav & Bethard 2018

NERC performance

Feature-engineered machine learning systems	Dict	SP	DU	EN	GE
Carreras et al. (2002) binary AdaBoost classifiers	Yes	81.39	77.05	-	-
Malouf (2002) - Maximum Entropy (ME) + features	Yes	73.66	68.08	-	-
Li et al. (2005) SVM with class weights	Yes	-	-	88.3	-
Passos et al. (2014) CRF	Yes	-	-	90.90	-
Ando and Zhang (2005a) Semi-supervised state of the art	No	-	-	89.31	75.27
Agerri and Rigau (2016)	Yes	**84.16**	**85.04**	**91.36**	**76.42**
Feature-inferring neural network word models					
Collobert et al. (2011) Vanilla NN +SLL / Conv-CRF	No	-	-	81.47	-
Huang et al. (2015) Bi-LSTM+CRF	No	-	-	84.26	-
Yan et al. (2016) Win-BiLSTM (English), FF (German) (Many fets)	Yes	-	-	88.91	**76.12**
Collobert et al. (2011) Conv-CRF (SENNA+Gazetteer)	Yes	-	-	89.59	-
Huang et al. (2015) Bi-LSTM+CRF+ (SENNA+Gazetteer)	Yes	-	-	**90.10**	-
Feature-inferring neural network character models					
Gillick et al. (2015) – BTS	No	**82.95**	**82.84**	**86.50**	**76.22**
Kuru et al. (2016) CharNER	No	82.18	79.36	84.52	70.12
Feature-inferring neural network word + character models					
Yang et al. (2017)	Yes	85.77	**85.19**	91.26	-
Luo (2015)	Yes	-	-	91.20	-
Chiu and Nichols (2015)	Yes	-	-	**91.62**	-
Ma and Hovy (2016)	No	-	-	91.21	-
Santos and Guimaraes (2015)	No	82.21	-	-	-
Lample et al. (2016)	No	85.75	81.74	90.94	**78.76**
Bharadwaj et al. (2016)	Yes	**85.81**	-	-	-
Dernoncourt et al. (2017)	No	-	-	90.5	-
Feature-inferring neural network word + character + affix models					
Re-implementation of Lample et al. (2016) (100 Epochs)	No	85.34	85.27	90.24	78.44
Yadav et al. (2018)(100 Epochs)	No	86.92	87.50	90.69	78.56
Yadav et al. (2018) (150 Epochs)	No	**87.26**	**87.54**	90.86	**79.01**

Table 1: Comparison of NER systemsin four languages: CoNLL 2002 Spanish (SP), CoNLL 2002 Dutch (DU), CoNLL 2003 English (EN), and CoNLL 2003 German (GE). Dict indicates whether or not the approach makes use of dictionary lookups. Best performance in each category is highlighted in bold.

Yadav, Vikas, and Steven Bethard. 2018.

- Feature-inferring NN models can outperform feature-engineered models
- Despite the latter's access to domain specific rules, knowledge, features, and lexicons

Pre-trained transformers fine-tuned for NERC

- Previous systems are trained only on annotated training data
- Pre-trained sentence-embeddings (transformers): weights (768 dimensions, 12 layers/attention heads) to model predictive value (MASKs) of tokens in sequence positions in sentence contexts:
    - “to run the show” versus "to run a marathon”
    - “Apple delayed the iPhone 13” versus “Apple with cinnamon go well together”
- Unsupervised pre-training on massive data
- Fine-tuned with small data set with (B)IO/IO(B) annotations, e.g., CoNLL2003

https://huggingface.co

The resource to find language models for text classification, QA, summarisation, many others

The AI community building the future.

Build, train and deploy state of the art models powered by the reference open source in machine learning.

Star 59,029

Tasks

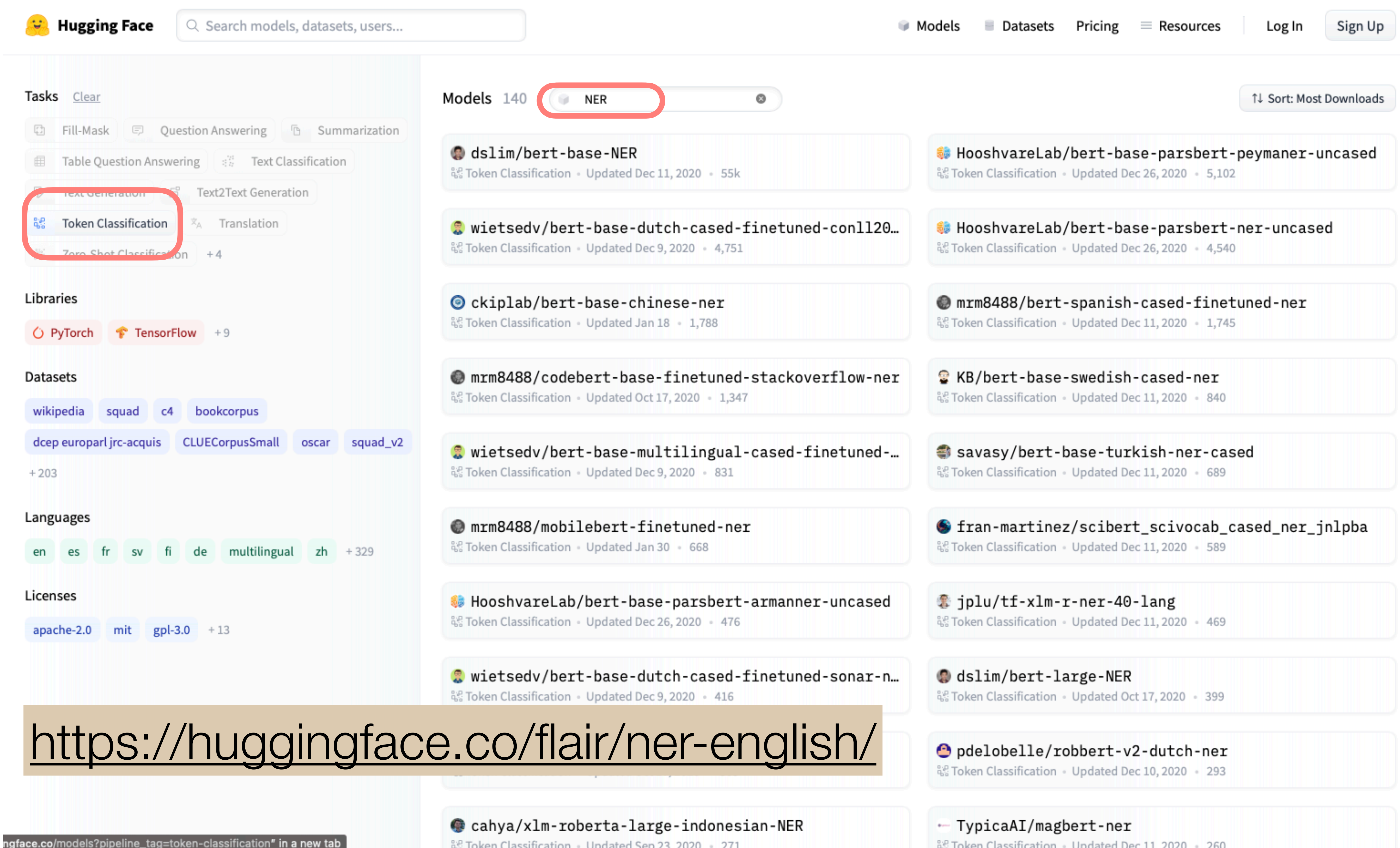
Problems solvers

Thousands of creators work as a community to solve Audio, Vision, and Language with AI.

Explore tasks

Task	Models
Text Classification	3471 models
Question Answering	1170 models
Translation	1469 models
Summarization	325 models
Audio Classification	66 models
Image Classification	278 models
Object Detection	15 models

Transformers fine-tuned for NERC
Token classification task: Sequence tagger



https://huggingface.co/flair/ner-english/

How to use fine-tuned transformer models

available tasks are ['audio-classification', 'automatic-speech-recognition', 'conversational', 'feature-extraction', 'fill-mask', 'image-classification', 'image-segmentation', 'ner', 'object-detection', 'question-answering', 'sentiment-analysis', 'summarization', 'table-question-answering', 'text-classification', 'text-generation', 'text2text-generation', 'token-classification', 'translation', 'zero-shot-classification', 'translation_XX_to_YY']"

In [1]:
```
from transformers import pipeline
```

In [2]:
```
nerc_classifier = pipeline("ner")
```

Large download

```
No model was supplied, defaulted to dbmdz/bert-large-cased-finetuned-conll03-english and revision f2482bf (https://huggingface.co/dbmdz/bert-large-cased-finetuned-conll03-english).
Using a pipeline without specifying a model name and revision in production is not recommended.
```

In [3]:
```
result = nerc_classifier("Hello I'm Ilia, I work at the VU University and I don't live in Weesp.")
```

In [4]:
```
print(result)
```

word-piece tokenisation

```
[{'entity': 'I-PER', 'score': 0.9959168, 'index': 5, 'word': 'Il', 'start': 10, 'end': 12}, {'entity': 'I-PER', 'score': 0.99507046, 'index': 6, 'word': '##ia', 'start': 12, 'end': 14}, {'entity': 'I-ORG', 'score': 0.9985386, 'index': 12, 'word': 'V', 'start': 30, 'end': 31}, {'entity': 'I-ORG', 'score': 0.9977562, 'index': 13, 'word': '##U', 'start': 31, 'end': 32}, {'entity': 'I-ORG', 'score': 0.987295, 'index': 14, 'word': 'University', 'start': 33, 'end': 43}, {'entity': 'I-LOC', 'score': 0.9819094, 'index': 22, 'word': 'We', 'start': 64, 'end': 66}, {'entity': 'I-LOC', 'score': 0.72425383, 'index': 23, 'word': '##es', 'start': 66, 'end': 68}, {'entity': 'I-LOC', 'score': 0.8684292, 'index': 24, 'word': '##p', 'start': 68, 'end': 69}]
```

In [5]:
```
sentiment_classifier = pipeline("sentiment-analysis")
```

```
No model was supplied, defaulted to distilbert-base-uncased-finetuned-sst-2-english and revision af0f99b (https://huggingface.co/distilbert-base-uncased-finetuned-sst-2-english).
Using a pipeline without specifying a model name and revision in production is not recommended.
```

In [6]:
```
result = sentiment_classifier("I really like Weesp. It is close to Amsterdam, has an old center and close to nature.")
```

In [7]:
```
print(result)
```

```
[{'label': 'POSITIVE', 'score': 0.998481810092926}]
```

Fine-tuned cross-lingual language models

Large download (takes a while)

In [8]:
```
nerc_classifier = pipeline("ner", model = "xlm-roberta-large-finetuned-conll03-english")
```

Downloading: 100% 2.24G/2.24G [12:47<00:00, 5.24MB/s]

```
huggingface/tokenizers: The current process just got forked, after parallelism has already been used. Disabling paral
lelism to avoid deadlocks...
To disable this warning, you can either:
        - Avoid using `tokenizers` before the fork if possible
        - Explicitly set the environment variable TOKENIZERS_PARALLELISM=(true | false)
```

Downloading: 100% 5.07M/5.07M [00:05<00:00, 1.34MB/s]

Downloading: 100% 9.10M/9.10M [00:06<00:00, 2.70MB/s]

In [9]:
```
result = nerc_classifier("Hallo ik ben Ilia, ik werk aan de Vrije Universiteit en ik woon niet in Weesp.")
```

In [10]:
```
print(result)
```

```
[{'entity': 'I-PER', 'score': 0.9998204, 'index': 4, 'word': '▁Ili', 'start': 13, 'end': 16}, {'entity': 'I-PER', 'sc
ore': 0.9998264, 'index': 5, 'word': 'a', 'start': 16, 'end': 17}, {'entity': 'I-ORG', 'score': 0.9999944, 'index': 1
1, 'word': '▁Vrij', 'start': 34, 'end': 38}, {'entity': 'I-ORG', 'score': 0.99999213, 'index': 12, 'word': 'e', 'star
t': 38, 'end': 39}, {'entity': 'I-ORG', 'score': 0.9999918, 'index': 13, 'word': '▁Universiteit', 'start': 40, 'end':
52}, {'entity': 'I-LOC', 'score': 0.999925, 'index': 19, 'word': '▁We', 'start': 72, 'end': 74}, {'entity': 'I-LOC',
'score': 0.99952745, 'index': 20, 'word': 'es', 'start': 74, 'end': 76}, {'entity': 'I-LOC', 'score': 0.9999051, 'ind
ex': 21, 'word': 'p', 'start': 76, 'end': 77}]
```

Factors that impact performance for NERC

- The annotation of the spans, annotation of nesting, ‘the’
  - [[[White House] [press] secretary] Scott McClellan]
  - [The [CEO] of the [US]-Based company [Facebook]]
- Genre of text: news or tweets/social media
- Entity types: people, organisations, amounts, dates, events (results differ per type)
- Amount of training data
- Difference between training data and test data:
  - domain dependency of entities
  - training data rapidly becomes obsolete

Domain specific NER

	Dict	MedLine (80.10%)			DrugBank (19.90%)			Complete dataset		
		P	R	F1	P	R	F1	P	R	F1
Feature-engineered machine learning systems										
Rocktäschel et al. (2013)	Yes	60.70	55.80	58.10	88.10	87.50	87.80	73.40	69.80	71.50
Liu et al. (2015) (baseline)	No	-	-	-	-	-	-	78.41	67.78	72.71
Liu et al. (2015) (MED. emb.)	No	-	-	-	-	-	-	82.70	69.68	75.63
Liu et al. (2015) (state of the art)	Yes	78.77	60.21	68.25	90.60	88.82	**89.70**	84.75	72.89	**78.37**
NN word model										
Chalapathy et al. (2016) (relaxed performance)	No	52.93	52.57	52.75	87.07	83.39	85.19	-	-	-
NN word + character model										
Yadav et al. (2018)	No	73	62	67	87	86	87	79	72	75
NN word + character + affix model										
Yadav et al. (2018)	No	74	64	**69**	89	86	87	81	74	77

91+ on CoNLL 2003

90+ on CoNLL 2003

Table 2: DrugNER results on the MedLine and DrugBank test data (80.10% and 19.90% of the test data, respectively). The Yadav et al. (2018) experiments report no decimal places because they were run after the end of shared task, and the official evaluation script outputs no decimal places.

Up to around 20 F1 points drop under cross-domain conditions.

Yadav & Bethard 2018

NERC References

- Yadav, Vikas, and Steven Bethard. "A survey on recent advances in named entity recognition from deep learning models." In Proceedings of the 27th International Conference on Computational Linguistics, pp. 2145-2158. 2018.

- Guillaume Lample, Miguel Ballesteros, Sandeep Subramanian, Kazuya Kawakami and Chris Dyer, 2016, Neural Architectures for Named Entity Recognition, NAACL.

- Zhiheng Huang, Wei Xu, Kai Yu 2015, Bidirectional LSTM-CRF Models for Sequence Tagging, arXiv.1508.01991v1

- Ilievski, Postma, and Vossen, Semantic overfitting: what `world' do we consider when evaluating disambiguation of text? COLING 2016.

- Rodrigo Agerri and German Rigau. 2016. Robust multilingual named entity recognition with shallow semi- supervised features. Artificial Intelligence, 238:63–82.

  - https://github.com/guillaumegenthial/tf_ner

  - https://www.depends-on-the-definition.com/named-entity-recognition-conditional-random-fields-python/

  - https://towardsdatascience.com/besides-word-embedding-why-you-need-to-know-character-embedding-6096a34a3b10

  - https://huggingface.co/course/chapter7/2?fw=pt

  - https://machinelearningmastery.com/develop-character-based-neural-language-model-keras/

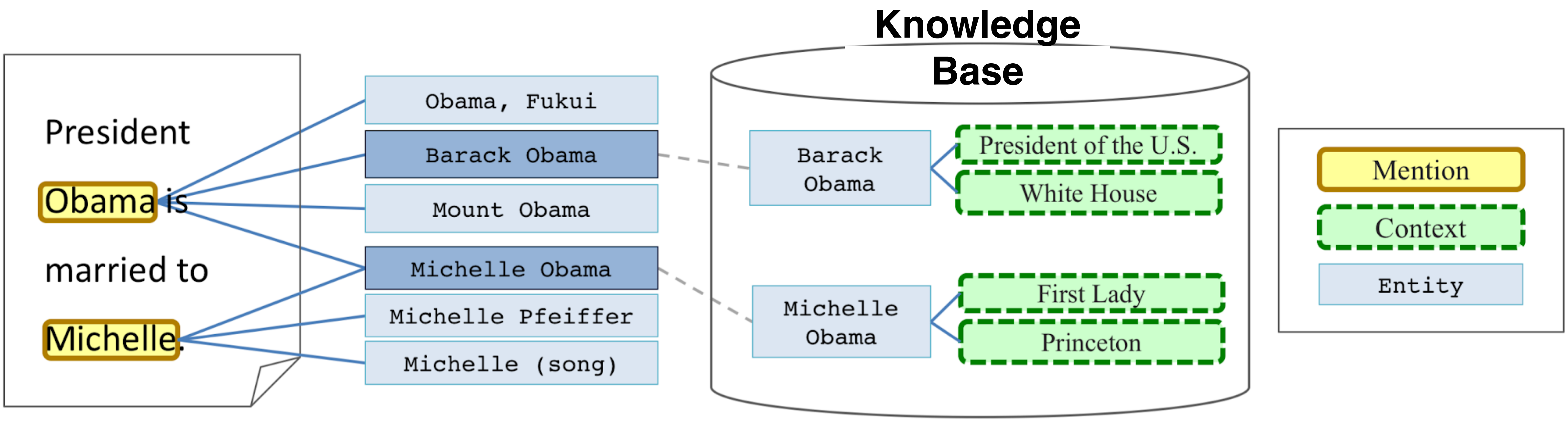
  - https://www.kaggle.com/abhinavwalia95/entity-annotated-corpus

Entity tasks in NLP

- NER (Recognition): detecting the phrase that is the name of an entity
- NEC (Classification): assigning an entity type to the phrase
- NEL (Linking): establishing the identity of the entity in a given reference database (Wikipedia, DBpedia, YAGO)
- Coreference: any phrase that makes reference to an entity instance, including pronouns, noun phrases, abbreviations, acronyms, etc…

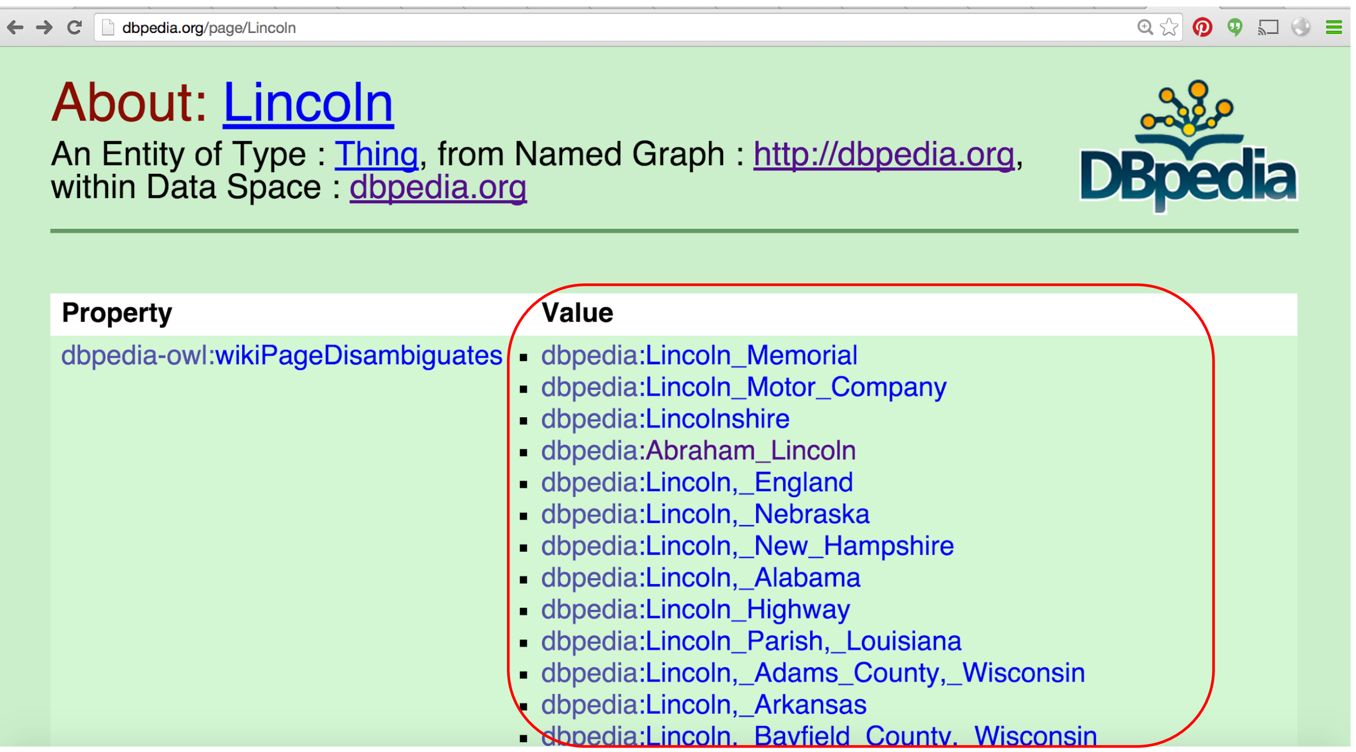
Entity Linking

(lab session 5)



- There are multiple entries in a database for Obama and Michelle.
- Based on the background knowledge in a database and the context (input data) ->
- Ranking of potential entities and identify the correct one

Named Entity Disambiguation isn’t that easy though



a. name **ambiguity**

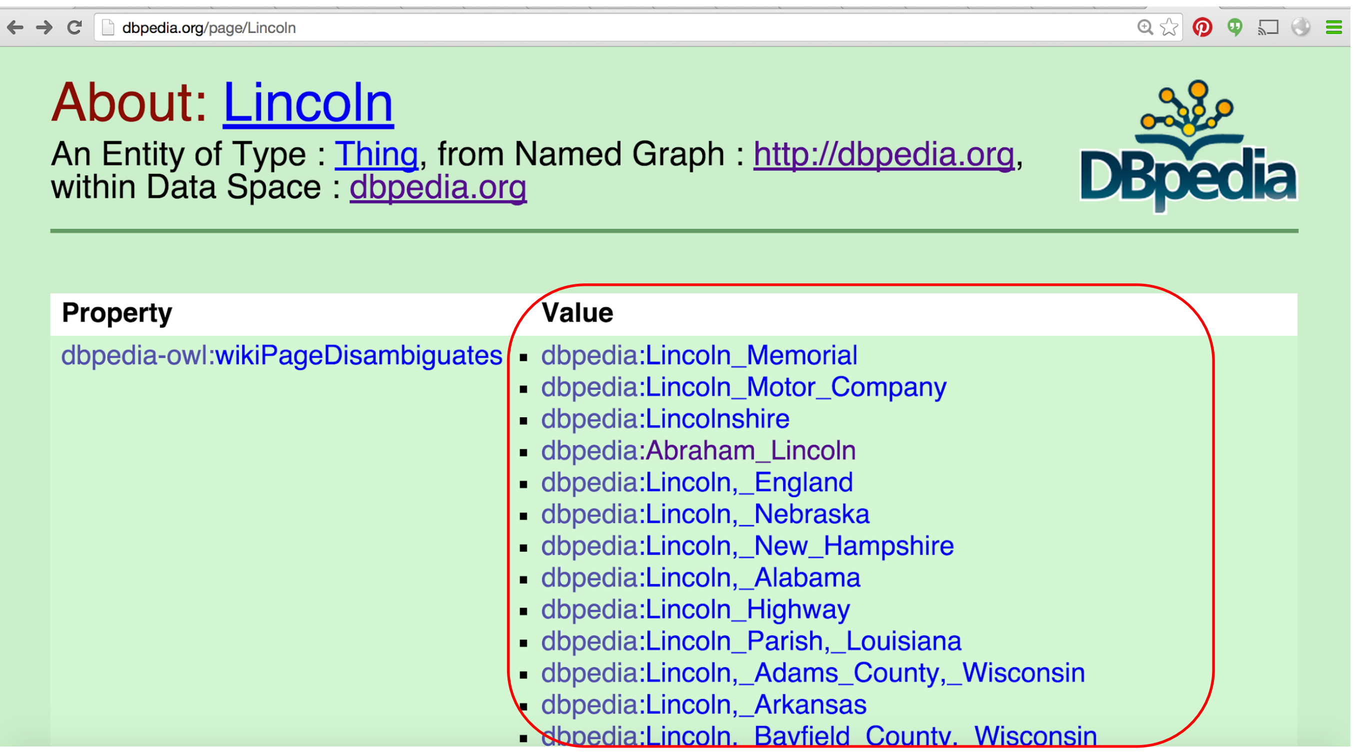
Very frequent => Wikipedia and DBpedia have special **disambiguation** pages that list the entities that are referred to by a mention.

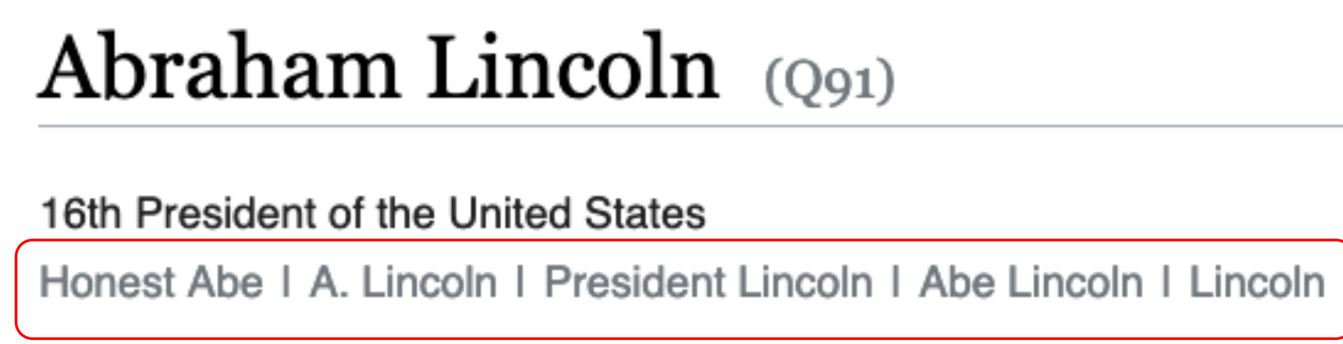
> *The more popular an entity the more vague our reference and the more ambiguous references get (polysemy)*

More populate NEs have more information in a database -> easier to resolve

Bias: systems are inclined to prefer the more popular interpretations

Named Entity Disambiguation isn't that easy though





a. name **ambiguity**

Very frequent => Wikipedia and DBpedia have special **disambiguation** pages that list the entities that are referred to by a mention.

b. name **variation**

(+ "Mr. Lincoln", "Lincoln", "Abraham", "Abe", "Honest Abe")

The more popular an entity the more references it will get and the more variation in reference (synonymy)

Missing (NIL) entities

February 27, 2019

SAN BENITO – Police have released more information connected to a murder investigation in San Benito.

Edgar Gonzalez, 30, was shot and killed last Thursday.

It happened on Buena Vida Street near the expressway and Sam Houston.

Police released a photo of a white Chevrolet Tahoe they say was used as a getaway vehicle.

Police are asking nearby businesses to check their surveillance video for any images of this vehicle.

If you have any information, contact the San Benito Police Department at 361-3880.

Problem of unpopular entities that may not be in a database

What to link “Edgar Gonzalez” to???

https://www.krgv.com/news/police-seeking-surveillance-footage-to-aid-in-san-benito-murder-investigation

We can look for the street in a database, we know the place is (San Benito), the brand of a car, but no Edgar Gonzalez

Missing (NIL) entities

From Wikipedia, the free encyclopedia

Edgar Gonzalez may refer to:

- Édgar González (pitcher) (born 1983), Mexican baseball pitcher
- Édgar González (Mexican footballer) (born 1980), Mexican football striker
- Édgar Daniel González (born 1979), Paraguayan international football player
- Edgar Gonzalez (infielder) (born 1978), American baseball infielder
- Edgar González (footballer, born 1997), Spanish footballer
- Edgar González, Jr. (born 1996), member of the Illinois House of Representatives

- There are a lot of Edgar Gonzalez mentioned in Wikipedia
- But is he really one of those, being shot at the age of 30 in February 27, 2019?

Alistair McAlpine: “trial by twitter”

- Lord McAlpine is a British Politician from the conservative party
- November 2012, massive outburst of tweets:
  - “Alistair McAlpine is a pedofile.“
  - “Arrested on suspicion of child abuse in Wrexham children's home”
- Wrong guy….. identity mismatch!
- Copied without fact checking by the British Press:
  - BBC settlement for £185,000
  - Claims up to £500,000 for ITV
- His lawyers identified >1k people that sent tweets and >9k people that retweeted messages

Entity linking methods

	Word-based	**Graph-based**
Main idea	Find the candidate with the most similar description to the one of a mention in text	Find candidates that are coherent with each other according to connections in the KB
Scoring example	measure text similarity, combine with **TF/IDF** weighting to measure relevance of a word	Put all candidates with their facts in a graph network and prune until only one candidate per mention is left
Decision unit	individual/local	collective/global
KB	unstructured (Wikipedia)	Structured (DBpedia, etc.)
Example	DBpedia Spotlight https://www.dbpedia-spotlight.org/demo/#	AIDA/AGDISTIS

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What is Coreference Resolution

- Coreference resolution is the task of finding out which words/phrases refer to the same entity

Abraham Lincoln Listeni/ˈeɪbrəhæm ˈlɪŋkən/ (February 12, 1809 – April 15, 1865) was the 16th President of the United States, serving from March 1861 until his assassination in April 1865. Lincoln led the United States through its Civil War—its bloodiest war and its greatest moral, constitutional and political crisis.[1][2] In doing so, he preserved the Union, abolished slavery, strengthened the federal government, and modernized the economy.

Lincoln grew up on the western frontier in Kentucky and Indiana. Largely self-educated, he became a lawyer in Illinois, a Whig Party leader, and a member of the Illinois House of Representatives, where he served from 1834 to 1846. Elected to the United States House of Representatives in 1846, Lincoln promoted rapid modernization of the economy through banks, tariffs, and railroads. Because he had originally agreed not to run for a second term in Congress, and his opposition to the Mexican–American War was unpopular among Illinois voters, Lincoln returned to Springfield and resumed his successful law practice. Reentering politics in 1854, he became a leader in building the new Republican Party, which had a statewide majority in Illinois. In 1858, while taking part in a series of highly publicized debates with his opponent and rival, Democrat Stephen A. Douglas, Lincoln spoke out against the expansion of slavery, but lost the U.S. Senate race to Douglas.

source: http://en.wikipedia.org/wiki/Abraham_Lincoln

But it's actually more complicated

- Coreference resolution is the task of finding out which words/phrases refer to the same object

Abraham Lincoln Listeni/ˈeɪbrəhæm ˈlɪŋkən/ (February 12, 1809 – April 15, 1865) was the 16th President of the United States, serving from March 1861 until his assassination in April 1865. Lincoln led the United States through its Civil War—its bloodiest war and its greatest moral, constitutional and political crisis.[1][2] In doing so, he preserved the Union, abolished slavery, strengthened the federal government, and modernized the economy.

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- We need to figure out that 'he', 'his', etc. refer to Lincoln and not to, e.g., Douglas.
- For Text Mining and Information Extraction, coreference resolution is an important technology.

source: http://en.wikipedia.org/wiki/Abraham_Lincoln